A Simple Mechanistic Model of Wave-Mean Flow Feedbacks, Poleward Jet Shifts and the Annular Mode

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ABSTRACT: Changes in the latitude of the zonal-mean mid-latitude jet play an important role for both natural variability, and the response of the atmospheric circulation to greenhouse gases and other external forcing. Nevertheless, the jet response to external forcing exhibits perplexing and non-intuitive behavior. For example, external forcing that acts to strengthen the jet will also shift the jet poleward. In addition, for internal jet variability, zonal wind anomalies slowly propagate poleward over most latitudes, however, this propagation is stalls somewhat at latitudes on the flanks of the mean jet. At these latitudes zonal wind anomalies are more stationary, and therefore anomaly persistence is maximized. These same persistent latitudes are colocated with the zonal wind anomalies associated with the annular mode. Feedbacks between the zonal-mean zonal wind and the eddy momentum fluxes are responsible for the above behaviors. Here a simple mechanistic model of the effect of the zonal-mean zonal wind on the eddy momentum fluxes is developed. The model reproduces the wave-mean flow feedbacks that maintain the annular mode, cause stronger jets to shift poleward (and vice versa), and cause the poleward propagation of zonal wind anomalies. In the model, the effect of the mean flow on the eddy momentum fluxes is determined solely by the critical level and the reflecting level. The model is used to distill the essential dynamics of annular variability and change such as why stronger jets shift poleward, why high frequency eddies are responsible for the positive feedback and why the intricate structure of propagating versus stationary zonal wind anomalies exists.

1. Introduction

The goal of this paper is to create the simplest mechanistic model that explains all of the main features of internal long-term variability of zonal-mean zonal wind (U) anomalies. For example, U anomalies propagate poleward with time (James and Dodd (1996); Feldstein (1998); Lee et al. (2007); Sparrow et al. (2009); Chemke and Kaspi (2015); Sheshadri and Plumb (2017)) and EOF1 U anomalies are reinforced by positive eddy momentum fluxes (Robinson (1991); Robinson (1996); Lorenz and Hartmann (2001); Lorenz and Hartmann (2003)). Furthermore, Lorenz (2015) showed that the latitudes of strongest poleward propagation are out of phase with EOF1 and that EOF1 U anomalies are relatively stationary in latitude with time. Lorenz (2015) also find that equatorward U propagation in the polar regions. Another robust feature of U variability is the fact that the high frequency eddies reinforce EOF1 U anomalies while the low frequency eddies damp U anomalies.

Robinson (2000) proposed a theory for the positive feedback and poleward propagation involving "the baroclinic mechanism": U anomalies cause colocated changes in the generation of wave activity via baroclinic instability, which then imply reinforcing eddy momentum flux convergence in the upper troposphere via the wave activity budget (Edmon et al. (1980)). This theory shows how upper-level potential vorticity (PV) fluxes can allow baroclinicity anomalies to be collocated with downgradient heat fluxes ($\propto$ baroclinic source of wave activity) despite the fact that the heat fluxes are acting to erode the baroclinicity anomalies. However, Robinson (2000) does not explain the structure of the EOF1 in relationship to the mean flow (Codron (2005)) and why the positive feedback is unique to EOF1. Also, while the effect of eddies on U is modelled quantitatively, effect of U on eddy fluxes is given by hypothetical prescribed structures. Therefore, the lack of a quantitative theory of the effect of U anomalies on eddy fluxes in Robinson (2000) renders the theory incomplete.

Jin et al. (2006a) and Jin et al. (2006b) developed a model that predicts annular mode structure from the mean state and the space-time structure of the transient eddy field. This model is barotropic, which suggests that eddy feedbacks do not depend on changes in baroclinic wave source (barotropic mechanism). Jin et al. (2006b) show how the meridional scale of the least-damped, leading mode depends on the spatial scale of the eddies and how the meridional scale of the low frequency modes affects the sign of the eddy feedback. However, this model does not consider poleward propagation. Moreover, because the model perturbs a prescribed "typical" eddy

field, interpretation in terms of the wave activity budget, critical levels and turning latitudes is difficult or impossible.

Lorenz (2015) developed a model that predicts eddy momentum fluxes from the mean state and the phase-speed/latitude structure of the baroclinic wave source (convergence of the vertical EP flux in the upper troposphere). Lorenz (2015) called the method Rossby Wave Chromatography (RWC) (Held and Phillips (1987)). RWC involves the forced linear barotropic vorticity equation, however, unlike other studies (Vallis et al. (2004); Chen et al. (2007); Barnes et al. (2010); Barnes and Hartmann (2011); Kidston and Vallis (2012)) the wave activity input is prescribed and the forcing is determined from the wave activity. The model reproduces the patterns of positive feedback and poleward and equatorward propagation in a GCM. Interpretation in terms of the wave activity budget is straightforward, and with careful analysis, the effect of critical levels and turning latitudes can be estimated (Lorenz (2014a); Lorenz (2014b)). Because the effect of the eddy momentum fluxes on the vertical average U is so simple (Lorenz and Hartmann (2001)), the eddy fluxes from RWC can be coupled to U and the evolution of the U anomalies to various forcings can be explored (Lorenz (2014b); Lorenz (2015)).

In this paper, we dramatically simplify the RWC model of Lorenz (2015): the linearized barotropic vorticity equation is simplified to just the critical level and the reflecting level (locus of turning latitudes). This simplification goes further than ray tracing because the full Index of Refraction (IOR) is collapsed to just the critical level (IOR = ∞) and the reflecting level (IOR = 0). In this simplified model the effect of critical and reflecting levels on the dynamics can be cleanly separated and the coupling between U and eddy momentum fluxes can be easily understood and quantified. Like Lorenz (2014b), we keep the amplitude of baroclinic wave generation fixed, which suggests that barotropic mechanisms are responsible for the patterns of positive feedback and propagation of U.

This paper begins with a description of the simple model and the application of the simple model to a GCM. We show that the simple model captures the main features of positive feedback and poleward and equatorward propagation in the GCM. Next the dynamical mechanisms that explain the model behavior are isolated. In particular, it is shown that reflecting level dynamics creates a specific latitude that attracts U anomalies and that critical level dynamics leads to U

anomaly propagation toward the baroclinic wave source. Finally, we show that similar patterns of U propagation and persistence occur in observations throughout the seasonal cycle.

2. Methods

a. GCM

The GCM is a standard primitive equation spectral model integrating the vorticity, divergence, temperature and the log surface pressure. The sigma coordinate vertical differencing scheme of Simmons and Burridge (1981) is used. An 8th order hyperdiffusion with a time scale of 0.1 days for the smallest scale waves is applied to the model variables. The only non-typical aspect of the model is the time differencing, which is the AB3-AI2 method of Durran and Blossey (2012). The resolution of all simulations is T85 with 20 equally spaced vertical levels. The model is forced with the diabatic heating and frictional damping of Held and Suarez (1994). The model is run for 6500 days and the first 500 days are discarded to allow for model spin-up.

The time mean zonal-mean zonal winds (U) with the control Held and Suarez (1994) parameters has a mid-latitude jet centered at about 42° latitude (Fig. 1a). Next, we calculate the EOFs of the instantaneous vertical- and zonal-mean zonal wind and then regress the zonal-mean wind on the resulting PCs (Fig. 1bc). EOF1 has oppositely signed center of actions on either side of the jet maximum and therefore represents north/south shifts in the mid-latitude jet. EOF2 is a tripolar pattern representing a strengthening and narrowing of the jet in its positive phase. In addition, a perturbation experiment is performed with the zonal-mean component of friction (Robinson (1997)) is reduced to 70% of its control value.

Space-time cross-spectral analyses of eddy fluxes are calculated using the method of Randel and Held (1991). The temporal spectral analysis is performed over 64 day chunks (= 192 times given the 8 hour sampling time) that overlap by 32 days. The resulting frequency spectrum is further smoothed with a running mean over five adjacent frequency bands. The phase speed spectra are given in terms of angular phase speed multiplied by $a\cos(45°)$, where a is the radius of the earth, so that the angular phase speed is approximately equal to the phase speed in the mid-latitudes. Typically, the stationary phase speed/frequency is not considered because the effects of strong stationary wave sources dominate the observed phase speed spectra. In our GCM with axisymmetric boundary

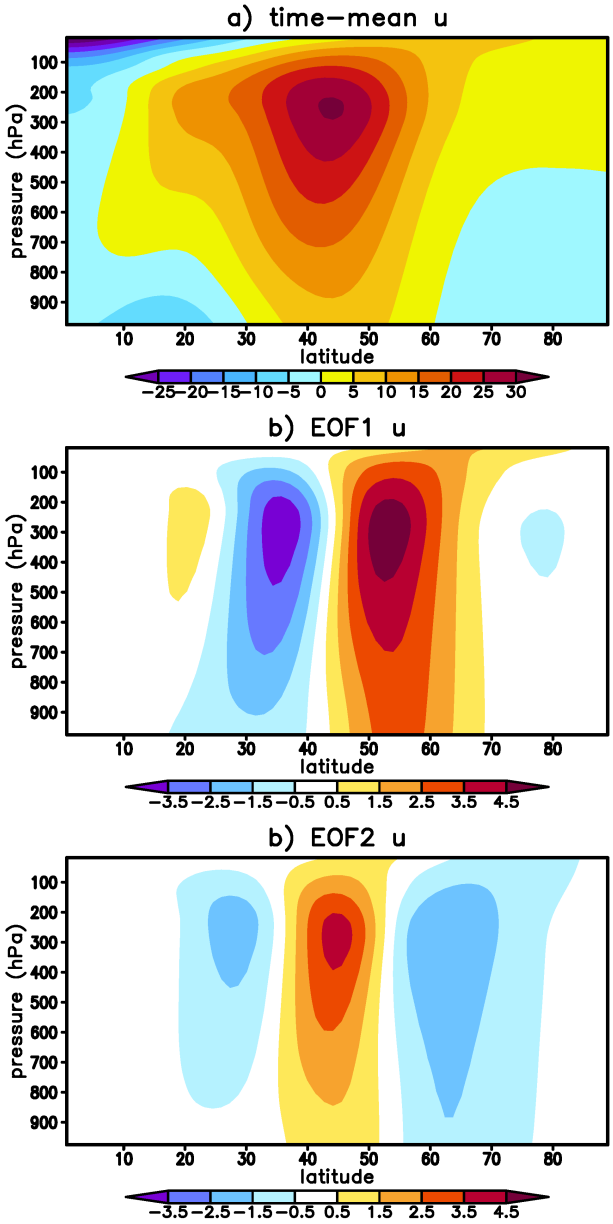

FIG. 1. a) Time- and zonal-mean zonal wind (U) from the control simulation (m/s). b) U anomalies regressed on PC1 of instantaneous U variability (m/s) c) Same as (b) but for PC2.

conditions, however, the stationary frequency (phase speed) is not enhanced relative to adjacent frequency bands and we therefore consider the full phase speed spectrum.

b. Simple Model Background

The simple model calculates the upper-level eddy momentum flux (UV = $\overline{u'v'}\cos(\phi)$[1], where ϕ is latitude) and the eddy momentum flux convergence (EMFC = $-1/\cos^2(\phi)\cdot\partial(\overline{u'v'}\cos^2(\phi))/\partial\phi$) from the baroclinic wave source (convergence of vertical component of the EP flux (Edmon et al. (1980)) in the upper troposphere) and the critical and reflecting levels (Lorenz (2014a); Lorenz (2014b)). The simple model is a simplification of the barotropic vorticity equation linearized about a steady, zonally uniform state. In this setting, waves preserve their zonal wavenumber (k) and angular phase speed (c = phase speed/$\cos(\phi)$). The mean U from the GCM depends on pressure. To convert to a single-level barotropic U, the full U is averaged vertically using weights taken from EOF1 of the eddy streamfunction (see Lorenz (2015)). The barotropic U defines the critical level:

$$\text{critical level} = \overline{u}/\cos(\phi). \tag{1}$$

According to linear Rossby wave theory, waves decay at the critical level. The barotropic U is also used to calculate the absolute vorticity gradient:

$$\beta^* = \frac{\partial(f+\overline{\zeta})}{a\partial\phi}, \tag{2}$$

where f is the Coriolis parameter and ζ is the relative vorticity ($\zeta = 1/(a\cos\phi)\cdot\partial(\cos\phi u)/\partial\phi$). The absolute vorticity gradient is used to calculate the reflecting level (see Lorenz (2014a) and Lorenz (2014b)):

$$\text{reflecting level} = \overline{u}/\cos(\phi) - \beta^* a^2\cos(\phi)/k^2, \tag{3}$$

where k is the integer zonal wavenumber. The reflecting level is the locus of Rossby wave turning latitudes and is equivalent to the Rossby wave dispersion relationship with the meridional wavenumber set to zero. For all figures, the critical and reflecting levels are scaled by $\cos(45°)$ so that these angular velocity based levels are approximately equal to the standard zonal wind in m/s in the mid-latitudes.

[1]Note, we define UV to be $\cos(\phi)$ times the true momentum flux, $\overline{u'v'}$. According to our definition the meridional wave activity flux is the negative of UV.

To best keep track of wave activity in the simple model, we scale the EP flux divergence (F, Edmon et al. (1980)) by $\cos(\phi)/a$:

$$\frac{\cos\phi}{a}\nabla\cdot F = \frac{-\partial(cos^2\phi\cdot\overline{u'v'})}{a\partial\phi} + \frac{\partial\left(f\cos^2\phi\cdot\overline{\theta'v'}/\Theta_p\right)}{\partial p}, \tag{4}$$

where p is the pressure, θ is the potential temperature and Θ_p is the derivative of the zonal-mean background θ. Writing the first term in terms of the EMFC, we have:

$$\frac{\cos\phi}{a}\nabla\cdot F = \cos^2\phi\cdot\mathrm{EMFC} + \frac{\partial\left(f\cos^2\phi\cdot\overline{\theta'v'}/\Theta_p\right)}{\partial p}. \tag{5}$$

The second term on the right is the negative of the baroclinic wave source:

$$\text{baroclinic wave source} = -\frac{\partial\left(f\cos^2\phi\cdot\overline{\theta'v'}/\Theta_p\right)}{\partial p}. \tag{6}$$

Because Θ_p is negative, the baroclinic wave source is positive in the upper troposphere under earth-like conditions. Finally, for time-mean conditions, the EP flux divergence is the dissipation of wave activity:

$$\text{dissipation} = \frac{\cos\phi}{a}\nabla\cdot F. \tag{7}$$

The dissipation is typically negative in the upper troposphere because the background potential vorticity gradient is positive. Rearranging the EP flux equation, we get:

$$\cos^2\phi\cdot\mathrm{EMFC} = \text{baroclinic wave source} + \text{dissipation}. \tag{8}$$

For the simple model, which uses one vertical level, the baroclinic source is integrated over the upper troposphere from 100 to 500mb. Hence, the EMFC caused by the wave source represents a vertical average over the same bounds. The integral of $\cos^2\phi\cdot\mathrm{EMFC}$ over latitude is identically zero.[2] If conditions at a certain latitude and phase speed, c, allow Rossby wave propagation, then $\cos^2\phi\cdot\mathrm{EMFC}$ is set equal to the baroclinic wave source at that latitude, which is positive, and the offsetting negative $\cos^2\phi\cdot\mathrm{EMFC}$ (= dissipation) occurs at the critical level for that c. If conditions

[2]Note that $\cos^2\phi\cdot\mathrm{EMFC}$ is the derivative of a quantity that is zero at the poles.

do not allow Rossby wave propagation, then the baroclinic wave source at that location makes zero contribution to the EMFC. More details of the simple model are described below after we first investigate the structure of the critical and reflecting levels.

c. Structure of Critical and Reflecting Levels

The critical level and reflecting level from the control run of the GCM are shown in Fig. 2. Except for a relatively minor angular velocity adjustment, the critical line is essentially the vertical- and zonal-mean zonal wind (solid green line in Fig. 2a). Because the reflecting line depends on the zonal wavenumber, the reflecting level is only shown for wavenumber 7 (dotted green line in Fig. 2a). The key feature of the reflecting level is its asymmetry about the latitude of maximum U: the reflecting level peaks on the poleward flank of the jet. To understand the reflecting level asymmetry consider the two components of the absolute vorticity gradient, β^* (equation 2): 1) the planetary component (β) and 2) a term that is essentially the negative of the second derivative of the zonal wind ($-\overline{u}_{yy}$). As expected for a smooth function, $-\overline{u}_{yy}$ is positive at the jet maximum with negative side lobes immediately to the north and south (red line in Fig. 2b). There is also a second relative maximum in $-\overline{u}_{yy}$ due to the slight U enhancement from the subtropical jet, but this is not important here. The planetary component slowly and monotonically decreases from equator to pole (blue line in Fig. 2b). When the two components are added together (purple line), the absolute vorticity gradient approaches zero only on the poleward flank of the jet because both the planetary and $-\overline{u}_{yy}$ components are small there. The smallness of the absolute vorticity gradient on the poleward flank means that the second term in the reflecting level equation (3) approaches zero and therefore the reflecting level approaches the critical level on the poleward flank of the jet.

d. Simple Model Description

The set of rules that govern the simple model are as follows:

- Suppose there is a baroclinic wave source at latitude, ϕ and phase speed, c. Waves propagate away from the baroclinic source (6) if c is greater than the reflecting level and less than the critical level. Otherwise, the baroclinic source does not contribute to the EMFC. These conditions on c are based on linear Rossby wave theory.

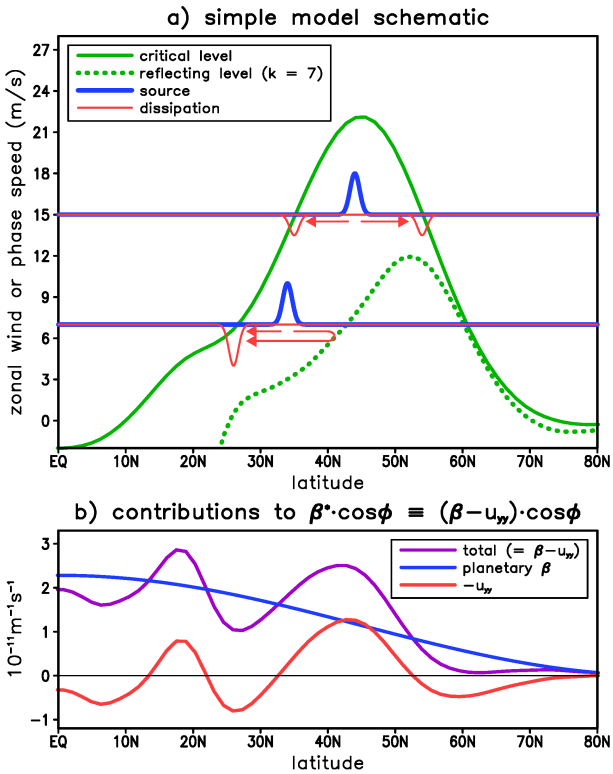

FIG. 2. a) The critical level (solid green) and reflecting level for wavenumber 7 (dashed green) from the control simulation. A schematic of a point source of baroclinic wave activity (blue) and the wave activity dissipation (red) for two different phase speeds. The red arrows show the path of the propagating waves. b) The zonal-mean barotropic PV gradient ($\beta^* \cos\phi$, purple) from the control simulation and the contribution of the planetary (blue) and zonal winds (red) to the total.

- If waves can propagate, then $\cos^2\phi \cdot$ EMFC at the source ϕ and c is set equal to the baroclinic source. In Fig. 2a the wave source is shown in blue for two separate values of ϕ and c.
- If waves can propagate, then half the waves propagate northward and half the wave propagate equatorward. This 50% split in wave direction is shown schematically with the red arrows in Fig. 2a. Waves preserve c when propagating, which is consistent with linear Rossby wave theory on a steady background flow.
- If the wave encounters a reflecting level, 100% of the wave activity reflects. This is shown schematically by the wave of lower phase speed in Fig. 2a. In this case, the wave activity propagating poleward reflects.

- If the wave encounters a critical level, the wave dissipates. This is shown by the red curves in Fig. 2a. Note that when there is reflection, all the dissipation occurs at a single critical line. Alternately, when there is no reflection, the dissipation is split 50/50 between the two critical levels surrounding the wave source.
- For the minor case of reflection both north and south of the wave source (trapped wave), the EMFC is set to zero. There are no trapped waves for the wavenumber 7 case shown.
- As given by wave activity budget (8), $\cos^2\phi \cdot$EMFC is the sum of the baroclinic wave source (blue) and dissipation (red). The quantity $\cos^2\phi \cdot \overline{u'v'}$ is found by integrating $-\cos^2\phi \cdot$EMFC from pole-to-pole with a boundary condition of zero. In the upper c case (Fig. 2a), UV is positive to the south of the source and negative to the north. In the lower c case, UV is positive (and larger than previous case) to the south of the source and zero to the north.

The baroclinic wave source is taken from the phase speed-wavenumber-latitude spectrum from the GCM (section 2a). Because of the discrete phase speeds in the spectrum and the fact that the wave dissipation is concentrated at the critical levels, there is excess noise in the EMFC in the vicinity of critical lines. The issue arises because all of the wave source is confined to a single c, when in reality the source is spread over the range $c-\delta/2c$ to $c+\delta/2c$, where δc is the grid spacing in c (= 1 /ms in our case). We correct this problem by uniformly partitioning the baroclinic wave source for a given c among n phase speeds from $c-\delta/2c$ to $c+\delta/2c$. Here we choose $n = 101$, which is probably excessive but the model is very fast so this is not a problem.

In addition, as currently formulated, the simple model assumes that when wave propagation is allowed, 100% of the wave activity propagates to the critical level. This leads to an overestimate of UV and the EMFC. Therefore, we assume that a fraction of the wave source does not propagate but dissipates in place. Let $f(\phi)$ be the total EMFC (i.e. integrated over c and k) from the simple model and $g(\phi)$ be the EMFC from the GCM. Then the fraction of wave activity that *does* propagate is set equal to $\sum_\phi f \cdot g / \sum_\phi f^2$. For the control simulation the empirical propagation "efficiency" is 54%, and for all figures below the UV and EMFC from the raw model are simply scaled by 0.54. This correction factor is consistent with EP flux diagrams (Edmon et al. (1980)), which show that the vertical EP flux dominates over the horizontal EP flux. This dominance of the vertical EP flux implies that a significant fraction of wave activity entering the upper troposphere dissipates locally rather than propagating meridionally.

When applying the simple model to perturbed U states, the phase speed spectrum of the baroclinic wave source is changed to account for the effect of advection and meridional vorticity gradients on the phase speed of baroclinic instability. For simplicity, we assume that the total wavenumber of the waves is the same as the zonal wavenumber (i.e. meridional wavenumber is zero). In this case, the phase speed of the waves is given by (3), and the phase speed change predicted by our method is:

$$\Delta c = \Delta\overline{u}/\cos(\phi) - \Delta\beta^* a^2 \cos(\phi)/k^2, \tag{9}$$

where the Δ refers to the value in the perturbed state minus the control. We have found that the actual wave source phase speed changes appear to be smoothed in latitude relative to predictions based on the dispersion relation. Therefore, the predictions in (9) are smoothed in latitude using a Gaussian kernel, $\exp(-\phi^2/a_c^2)$, with $a_c = 5°$. The predicted phase speed change is then used to shift the baroclinic wave source spectrum of the control run in a conservative way by mapping the power in each phase speed bin to the two bins closest to the new predicted phase speed. When exploring detailed mechanisms below, the results without the source phase speed change will also be considered. Also, the amplitude of the wave source at each latitude is unchanged in all cases. In other words, there is zero baroclinic feedback in this paper.

Finally, when the EMFC is used to infer the amount of positive feedback or poleward shift, we multiply the EMFC by 4/9 to convert upper tropospheric EMFC (100-500mb) to vertical average EMFC (100-1000mb). (The ratio of the thickness of the 100-500mb and 100-1000mb layers is 4/9.)

e. Simple Model Example

First we show the EMFC calculated from the simple model for the control GCM simulation (Fig. 3). Because the reflecting level depends on the zonal wavenumber (k), we show results for a single k ($k = 7$). The critical level, reflecting level and baroclinic wave source (Fig. 3a) are the prescribed inputs to the simple model. According to the simple model, wave propagation only occurs for wave sources with phase speeds between the reflecting and critical levels, therefore positive EMFC is subject to the same restriction (Fig. 3). By construction, all negative EMFC is confined to the critical level. Despite the fact that initial wave propagation is equally partitioned between poleward and equatorward propagating waves, the fact that the reflecting level peaks on the poleward flank

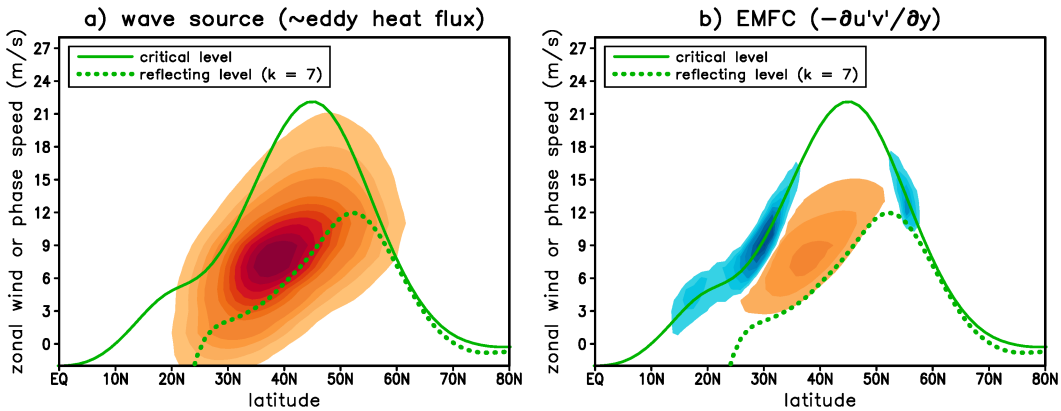

FIG. 3. Baroclinic wave source (shaded) for wavenumber 7 from the control simulation. The critical (solid green) and reflecting (dashed green) levels are also shown. b) EMFC$\cdot\cos\phi$ for wavenumber 7 calculated by the simple model (shaded).

of the jet causes most of the negative EMFC to be concentrated at the subtropical critical level (recall that waves preserve phase speed, c, in the simple model).

f. Jet Shift and Persistence Diagnostics

To explore the poleward propagation and persistence of the internal variability of the GCM, we perform a one-point lag regression analysis on U anomalies for all latitudes (Feldstein (1998)). Because the GCM has weak subtropical jets, the vertical average U is used. For the quantitative diagnostics of shift and persistence as a function of the base latitude of lagged regression, we perform one point lagged regressions on pentad data. The methodology is taken from Lorenz (2015). Briefly, let $u(\phi,t)$ be the latitudinal profile of U at lag t. We find the amplitude scaling, A, and latitudinal shift, ϕ_0, that minimize the difference between the lag 0 and lag t U profiles:

$$\min \sum_{\phi} \left(u(\phi,t) - Au(\phi-\phi_0,0)\right)^2 . \tag{10}$$

For the results here, $t = 2$ pentads.

For quantifying the shift and persistence in the simple model, we compute the change in EMFC in response to an imposed U perturbation. Next we project the EMFC on either U itself (positive

feedback) or on $-\frac{\partial u}{\partial \phi}$ (poleward shift, see Kushner et al. (2001)) and then normalize:

$$b = \frac{\sum_\phi \overline{u} \cdot \mathrm{EMFC}}{\sum_\phi \overline{u}^2},$$
$$p = \frac{\sum_\phi -\frac{\partial \overline{u}}{\partial \phi} \cdot \mathrm{EMFC}}{\sum_\phi \left(-\frac{\partial \overline{u}}{\partial \phi}\right)^2}, \tag{11}$$

where b is the positive feedback parameter and p is the poleward shift parameter, which specifically is the latitude shift per time caused by the imposed U perturbation. As shown in Lorenz and Hartmann (2001), the U tendency equation is very well approximated by EMFC plus Rayleigh damping:

$$\frac{d\overline{u}}{dt} = \mathrm{EMFC} - D\overline{u}, \tag{12}$$

where D is the Rayleigh damping coeficient. For the positive feedback component, EMFC = $b\overline{u}$, where b is a positive constant, and therefore:

$$\frac{d\overline{u}}{dt} = (b - D)\overline{u}, \tag{13}$$

where D $(= 0.125 day^{-1})$ is the Rayleigh damping coefficient estimated from the control run using the method of Lorenz and Hartmann (2001). Solving (13) gives an amplitude scaling of $\exp((b - D)t)$ relative to the initial state. Lorenz (2015) show that the GCM lagged regressions have a transient period of negative EMFC that is forced by the initial U perturbation. Since the simple model is a steady state model, we must account for this decay separately. Therefore, for all U anomalies, we multiply the amplitude by 0.8 to account for the transient decay (Lorenz (2015)):

$$A = 0.8 \exp((b - D)t). \tag{14}$$

If (12) is assumed to apply at all latitudes, the poleward shift tendency equation also has Rayleigh damping and the EMFC is p times the relative amplitude of the current U anomaly ($\propto \exp((b - D)t)$):

$$\frac{d\Delta\phi}{dt} = -D\Delta\phi + p \exp((b - D)t), \tag{15}$$

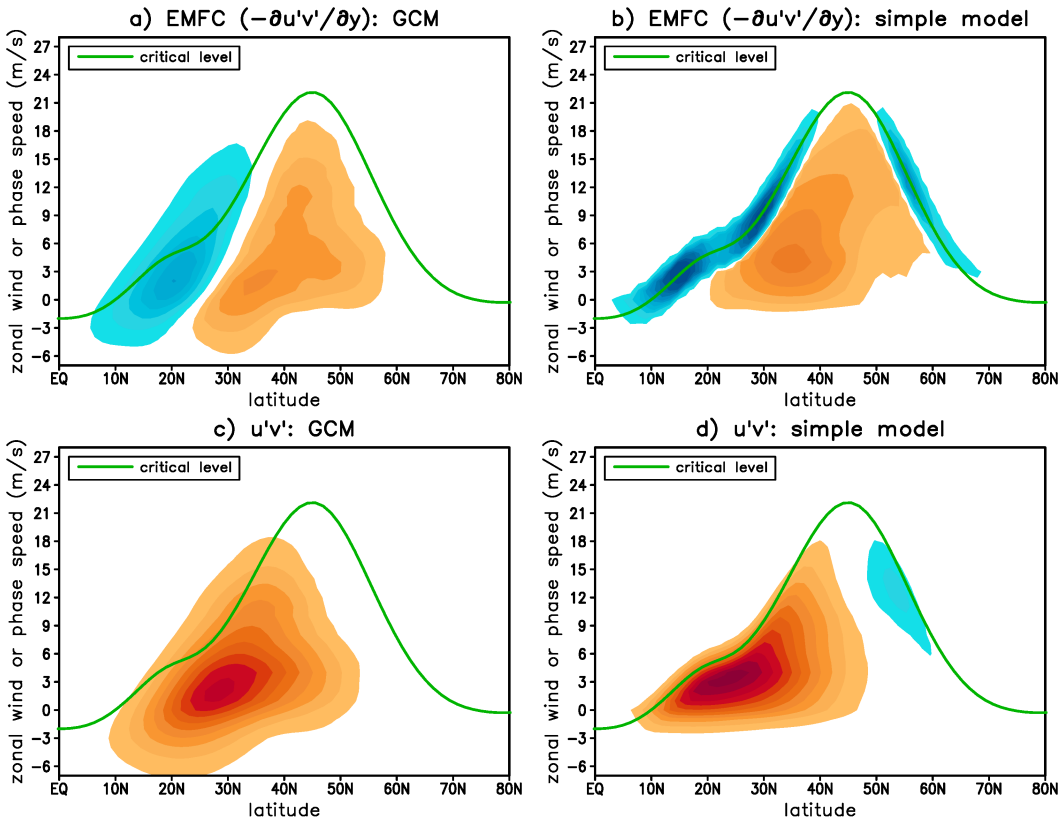

FIG. 4. a) EMFC$\cdot\cos\phi$ summed over all wavenumbers for the control simulation of the GCM. Contour interval = 0.04 m/s/day. The green line is the critical level. b) Same as (a) but for the simple model. c) $\overline{u'v'}\cdot\cos\phi$ (= UV) summed over all wavenumbers for the control simulation of the GCM. Contour interval = 0.4 m^2/s^2. d) Same as (c) but for the simple model.

where $\Delta\phi$ is the poleward shift, the first term on the right is the Rayleigh damping of the poleward shift and the second term is the poleward shift response caused by the initial U perturbation, which is decaying in time. The solution to (15) with initial condition zero is the poleward shift in the simple model:

$$\Delta\phi = p\frac{\exp(bt)-1}{b}\exp(-Dt). \tag{16}$$

In summary, the lag 0 one-point-regression U profile from the GCM is applied as a perturbation to the simple model to calculate the EMFC. The EMFC is projected on the U perturbation itself and the poleward propagation pattern using (11). Finally, (14) and (16) are used to find the amplitude and shift after 10 days. This is then compared to the values in the GCM calculated from (10).

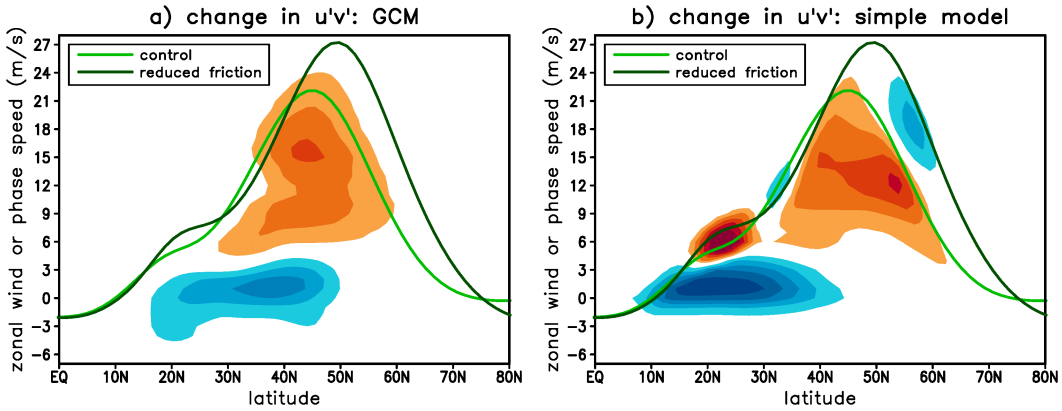

FIG. 5. a) Change in $\overline{u'v'} \cdot \cos\phi$ (= UV) in response to reduced zonal-mean friction in GCM. Contour interval = 0.3 m^2/s^2. The critical level for the control (light green) and reduced friction (dark green) are also shown. b) Same as (a) but for the simple model.

3. Results

a. Comparison of Simple Model and GCM

A comparison between the GCM and the simple model for EMFC summed over all zonal wavenumbers is shown in Fig. 4ab. The positive EMFC shows the best agreement in the general amplitude and structure. For the negative EMFC, it is apparent that the assumption that all wave dissipation occurs at the critical level is a bit too extreme. In addition, the amount of wave dissipation on the poleward flank of the jet is larger in the simple model, although this bias is exaggerated in the figure by the concentration of negative EMFC at the critical level in the simple model. Because UV is the integral of the EMFC, it emphasizes the large scale features and is less impacted by the wave dissipation assumptions of the simple model (Fig. 4cd). In this field, the bias in poleward wave propagation (= negative UV) is much less evident.

In Fig. 5, we compare the response to decreased zonal-mean friction. First note that, as expected, the zonal-mean zonal winds (= critical level) have increased due to decreased surface friction (compare the light and dark green lines in Fig. 5a). In addition, the latitude of the jet peak has shifted poleward (Robinson (1997)). The change in UV is positive at high phase speeds and negative at low phase speeds. When integrated over phase speed, the positive UV dominates at most latitudes. The positive UV across the mean jet gives positive EMFC on the poleward flank of the jet, causing the poleward shift in U. The simple model does a good job reproducing the two

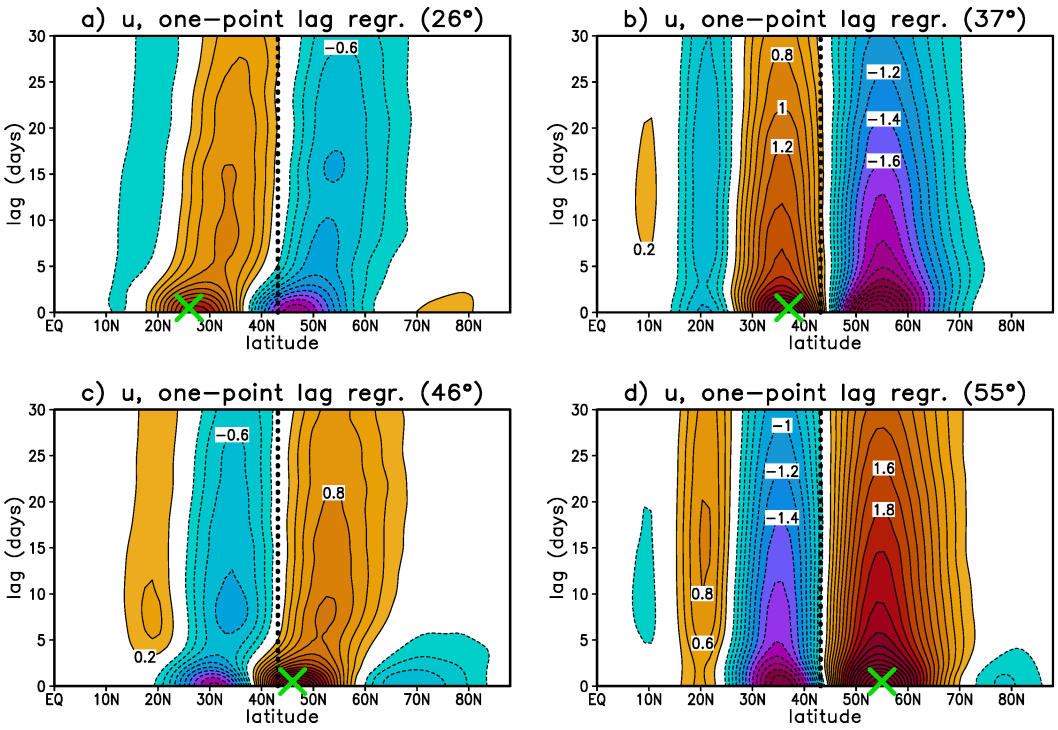

FIG. 6. a) One-point-lagged regression of vertical-average U anomalies on the vertical-average U at 26° latitude from the GCM (m/s). The green X at lag 0 denotes the base latitude. The thick, dotted vertical line denotes the latitude of the time-mean jet. b) Same as (a) but for base point 37°. c) Same as (a) but for base point 46°. d) Same as (a) but for base point 55°.

main UV structures seen in the GCM (Fig. 5b). However, the simple model has additional small scale structures in the vicinity of critical levels.

To diagnose the structure of poleward propagating versus stationary anomalies in the GCM, we calculate one-point lagged regression plots of the vertical- and zonal-mean zonal wind (see section 2f). For some base points, such as 26° and 46° latitude, U anomalies propagate poleward in time (Fig. 6ac). For others, such as 37° and 55° latitude, U anomalies remain stationary in time (Fig. 6bd). In addition, the stationary anomalies appear to decay less than the propagating anomalies and they have a nodal line that is almost collocated with the time mean jet (black dotted line in Fig. 6). The stationary anomalies are also in phase with EOF1 (Fig. 1b). The propagating anomalies tend to evolve towards an EOF1 state with a nodal line almost collocated with the time mean jet.

To summarize the one-point lagged regression results for all base points, we calculate the persistence and degree of poleward propagation for each base point. The persistence is defined as

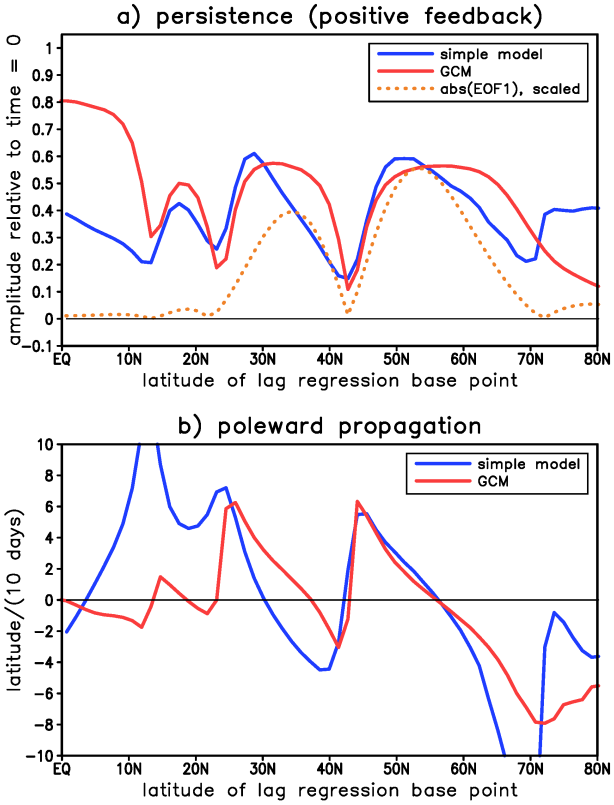

FIG. 7. a) Persistence of one-point-lag regression U structures at time = 10 days as a function of base point for the GCM (red) and the simple model (blue). For reference the absolute value of EOF1 is shown (dashed orange, arbitrarily scaled for comparison purposes). b) As in (a) but for poleward propagation.

the the ratio of the amplitude at time lag = 10 days relative to the amplitude at time = 0, and the poleward propagation is defined as the latitudinal shift at time 10 days compared to time = 0 (see section 2f for more details). There are four distinct zones of relatively high persistence (red line in Fig. 7a). The two zones in the extratropics are collocated with the centers of action of EOF1 (the scaled absolute value of EOF1 is given by the dotted orange line). The poleward propagation in the GCM maximizes in two distinct regions: the subtropics (26°) and the mid-latitudes (46°) (red line in Fig. 7b). In the polar regions, equatorward propagation dominates, which, given the poleward propagation further equatorward, implies that U anomalies are "attracted" towards latitudes around 55°.

To compare the simple model dynamics to the one-point lagged regression, we add the lag 0 U anomalies associated with every base point to the time-mean U and calculate the EMFC response. Next we use the EMFC response to integrate the U anomalies forward in time for 10 days (see

section 2f for more details). In the subtropics and the mid-latitudes, the simple model does a reasonable job capturing the amplitude and the structure of the persistence (Fig. 7a). The poleward propagation is also reasonable in these same regions, however, the simple model has an equatorward propagation bias from latitudes 25° to 40°. In the deep tropics and in the far polar regions, the simple model is significantly worse, however, U anomalies are weaker here and therefore less important. Overall, however, the simple model captures the persistance and poleward propagation of the GCM in the mid-latitudes and we believe that the dynamical processes operating in the GCM can be diagnosed from the simple model.

b. Understanding the Simple Model

The results the previous section (Fig. 7) involve U anomalies from the GCM, which vary in structure depending on the basepoint of the one point lagged correlation. To avoid this complicating factor, we perform a series of experiments with a Gaussian shaped U perturbation of the form $2\exp(-\phi^2/a_u^2)$, where $a_u = 11°$. The spatial scale of this U perturbation is about the same as the poleward center of action of EOF1. To gauge the positive feedback of the EMFC back on the imposed U perturbation, we project the EMFC from the simple model back on the imposed U. To gauge the effect of EMFC on the latitude of the imposed U, we project the EMFC on the $-\partial\overline{u}/\partial\phi$. For small perturbations, this quantifies the amount of poleward propagation because by a Taylor approximation the anomalies associated with a poleward shift of δ are $\overline{u}(\phi-\delta)-\overline{u}(\phi) \approx \delta\cdot(-\partial\overline{u}/\partial\phi)$ (Kushner et al. (2001)).

In Fig. 8, the positive feedback and poleward propagation are shown as a function of the latitude of the imposed U anomaly. Because the shape of U is consistent across all experiments, the structure of the curves is much smoother than Fig. 7, especially for the poleward propagation. There are two main locations of positive feedback in the mid-latitudes (Fig. 8a), which are located on the flanks of the mean jet (red dotted line) and are approximately colocated with the centers of action of EOF1 (purple dotted line). The simple model does not perform as well near the pole so the large feedback poleward of 70° is probably not dynamically relevant. There are two distinct regions of poleward propagation (Fig. 8b): a strong, broad region in the subtropics and tropics and a relatively weak and narrower region at the jet peak and slightly poleward. Poleward

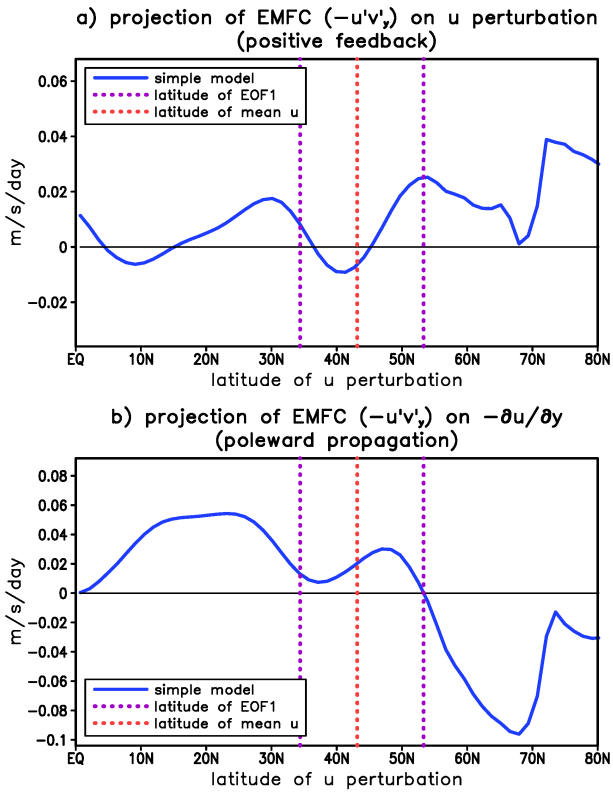

FIG. 8. a) Positive feedback of the EMFC from the perturbed simple model onto the imposed Gaussian U anomaly (blue). The latitudes of the time-mean jet (red) and the centers of action of EOF1 (purple) are also shown. b) Same as (a) but for the poleward propagation.

of 54°, equatorward propagation dominates. The juxtaposition of the poleward and equatorward propagation in the vicinity of 54° means that U anomalies are "attracted" towards 54° in time.

To better understand the above phenomena, we classify the reasons for EMFC changes in the simple model into the following categories:

- The phase speed of the baroclinic wave source changes (9).
- The ability to propagate changes: c is between the critical and reflecting level in either the control or the perturbed state but not both.
- The number of reflections does not change, therefore since items 1 and 2 have already been excluded, the change must be from a change in the location of the critical level. This is denoted "the critical level" case below.
- The number of reflections changes from 0 to 1 or vice versa. This is denoted "the 0-1 reflection" case below.

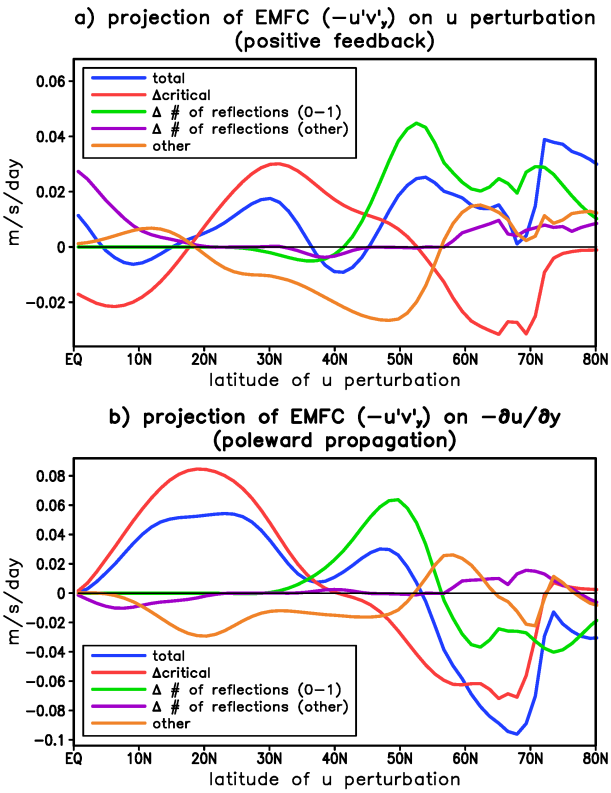

FIG. 9. As in Figure 8 except that the total response (blue) is partitioned into different mechanisms (see text).

- The number of reflections changes from 0 to 2, 1 to 2 or vice versa.

The first two items (phase speed and ability to propagate) are not negligible but they do not contribute to the features discussed above, instead they tend to act opposite the total EMFC response. Therefore, we combine these two mechanisms into a single "other" category. The contributions of the above categories to the positive feedback and the poleward shift are shown in Fig. 9 together with the total (blue). For the positive feedback, the critical level (red) is responsible for the feedback equatorward of 42° and the change in reflection from 0-1 (green) is responsible for the positive feedback poleward of 42°. All other categories work against the total positive feedback in the mid-latitudes. Similarly, for poleward propagation only the critical level and the 0-1 reflection contribute positively to the total response. With the classification, we see that the two distinct regions of poleward propagation are caused by two distinct processes: the subtropical region is cause by changes in the critical level and the mid-latitude region is caused by 0-1 reflection. These results are consistent with Lorenz (2014b) but the separation of the processes is much simpler and cleaner with the simple model. In the following two subsections we discuss

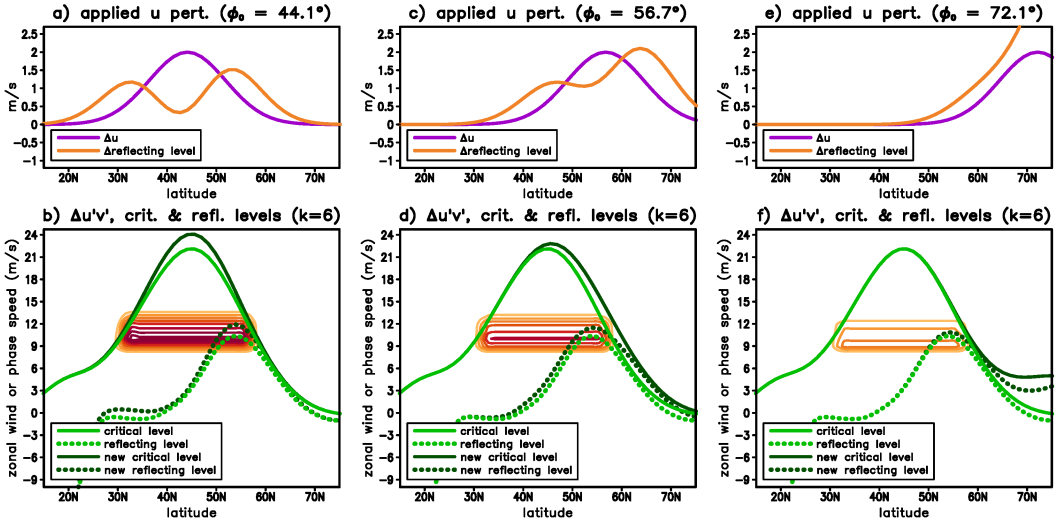

FIG. 10. a) Change in U (purple) and reflecting level (orange) from a Gaussian U perturbation centered at latitude 44.1°. b) Change in $\overline{u'v'}\cos^2\phi$ for wavenumber 6 in response to a Gaussian U perturbation at 44.1° (orange/red contours). The critical (solid) and reflecting (dashed) levels for the control (light green) and perturbed (dark green) are also shown. c) Same as (a) but for U perturbation at 56.7°. d) Same as (b) but for U perturbation at 56.7°. e) Same as (a) but for U perturbation at 72.1°. f) Same as (b) but for U perturbation at 72.1°.

in more detail the two cases that give rise to the interesting structures in the positive feedback and the poleward propagation: the 0-1 reflection case and the critical level case.

c. *Reflection Mechanism*

In Fig. 10, we look in detail at the 0-1 reflection for zonal wavenumber 6 for a few different U perturbations. First, the change in U and the reflecting level for the U perturbation centered at $\phi_0 = 44.1°$ is shown (Fig. 10a). The reflecting level change is a combination of U itself and a term approximately proportional to $-u_{yy}$. The $-u_{yy}$ nearly cancels the effect of U perturbation at ϕ_0 but reinforces the reflecting level increase on the flanks (orange line). The key change for the wave fluxes is the fact that the peak of the reflecting level extends to higher phase speeds (compare light and dark green dotted lines in Fig. 10b). This means that waves that were once absorbed on polar flank of jet are now reflected, leading to anomalous equatorward wave activity flux across the jet from critical level to critical level. Since the momentum flux is opposite the wave activity flux, this means anomalous poleward momentum flux (orange/red contours in Fig. 10b). Also note that the change in UV in confined to the narrow range of phase speeds spanned by the old and new

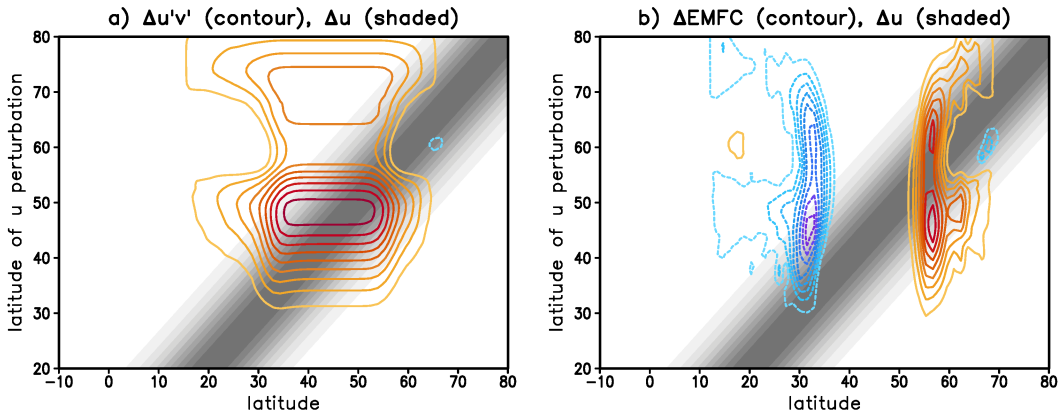

FIG. 11. a) Change in 0-1 reflection $\overline{u'v'}\cos^2\phi$ (contours) as a function of latitude (x-axis) and latitude of imposed U perturbation (y-axis). $\overline{u'v'}\cos^2\phi$ is integrated over all c and k. The imposed U perturbation is shaded gray. b) Same as (a) but for EMFC$\cdot\cos^2\phi$ instead of $\overline{u'v'}\cos^2\phi$.

peak of the reflecting level. A similar pair of plots is shown for a U perturbation at $\phi_0 = 56.7°$ (Fig. 10cd). Due to various spherical geometry terms, the relationship between the change in U and the reflecting level in Fig. 10c is somewhat different, but the reflecting level still increases at all latitudes near the U perturbation. Therefore, the peak of the mean reflecting level still increases and positive UV anomalies appear across the jet (Fig. 10d). Finally, a U perturbation as far north as 72.1° increases the peak of the reflecting level slightly, leading to the exact same pattern of UV increases across the jet (Fig. 10f). The key point is that wide range of U anomalies spanning from approximately 35 to 75° lead to essentially the same shape UV anomaly. We highlight the implications of this fact in the next paragraph. Also, note that the change in PV gradient ($-u_{yy}$) plays an important role in the reflecting level changes especially for $\phi_0 = 44.1°$. This suggests that the zonal wind "overriding" method of Chen et al. (2020) may miss aspects of the reflecting level response.

For the total UV, the latitude of the peak of the reflecting level varies with zonal wavenumber, however, the peak is does not vary enough to override the results above. For example, Fig. 11a shows the 0-1 reflection UV for all waves (x-axis) as a function of the latitude of the U perturbation (y-axis). The imposed U perturbation is shaded gray for reference. There are some differences in structure, especially for U anomalies near 60°, but they are very minor. Note also that the UV amplitude varies in response to the double peak structure of the reflecting level change (see Fig. 10a, orange line). The EMFC associated with the UV is shown in Fig. 11b. Note that the

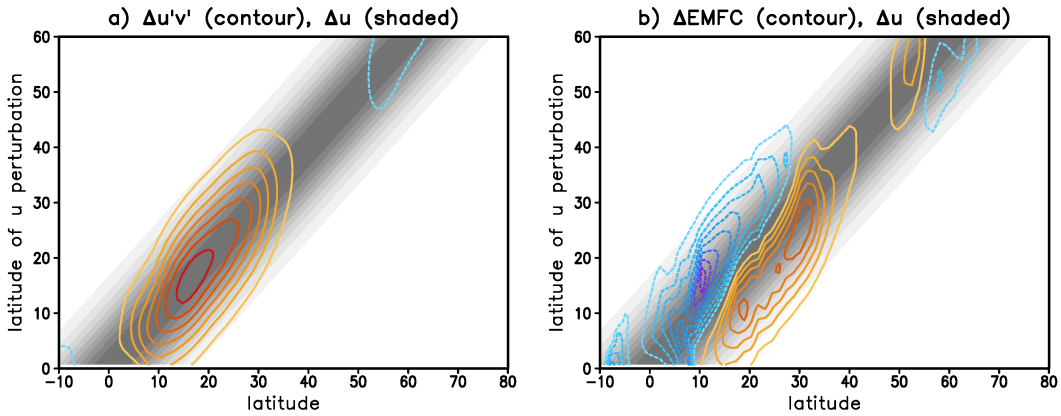

FIG. 12. Same as Figure 11 but for the eddy response from critical levels instead of 0-1 reflection.

positive EMFC is concentrated around 57° for wide range of positive U anomalies (and vice versa for negative U and EMFC). This implies that 57° is an attractor for U anomalies because nearby U anomalies produce an eddy response that drives the U toward 57°[3]. This attractor is why 1) stronger jets shift poleward (i.e. U anomaly at 42° propagates toward 57°), 2) a positive feedback exists for the poleward center of action of EOF1, and 3) why equatorward U propagation exists in the polar regions. Such an attractor occurs whenever the EMFC response is insensitive to the location of the imposed U anomaly.

d. Critical Level Mechanism

The UV and EMFC response (x-axis) as a function of the latitude of the U perturbation (y-axis) for the critical level case is shown in Fig. 12. Unlike the 0-1 reflection case, the UV pattern (contours) tends to track the location of the U perturbation (shaded), although there is a hint of deviation for U anomalies in the deep tropics and U anomalies near 40°. There are also weak negative UV for U anomalies at 50-60°. The EMFC (Fig. 12b) tends to be out of phase with the imposed U anomaly and therefore acts to shift the U in latitude: poleward shifts in the subtropics and equatorward shifts further poleward. This is consistent with the red curve in Fig. 9b.

The dynamics of the critical level effect can be seen by looking at the pattern of UV response for individual U perturbations (Fig. 13). The U perturbation (purple) and the total UV response over all wavenumbers (red) is shown in the top panels, and the old and new critical level and the ΔUV phase speed spectrum are shown in the bottom panels. In response to the increase in U, the

[3]Note that the equatorward propagation due to critical levels in this region slightly shifts the location of attractor latitude to be 54° instead of 57° (compare zero crossing of green and blue lines in Fig. 9b)

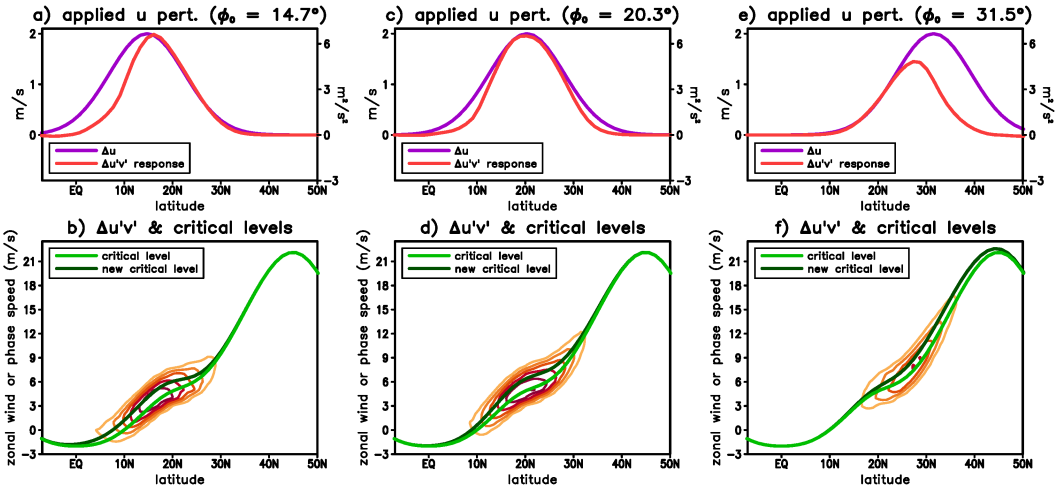

FIG. 13. a) Change in U (purple) and $\overline{u'v'}$ (red) from a Gaussian U perturbation centered at latitude 14.7°. b) Change in $\overline{u'v'}$ (integrated over k) in response to a Gaussian U perturbation at 14.7° (orange/red contours). The critical levels for the control (light green) and perturbed (dark green) are also shown. c) Same as (a) but for U perturbation at 20.3°. d) Same as (b) but for U perturbation at 20.3°. e) Same as (a) but for U perturbation at 31.5°. f) Same as (b) but for U perturbation at 31.5°.

critical level moves to lower latitudes and therefore there is anomalous equatorward wave activity flux between the old and new critical levels. Because UV is opposite the wave activity flux, this implies poleward momentum flux anomalies. Because the critical level changes are colocated with the anomalous UV, the U and UV are in phase, which means the EMFC is out of phase and causes a poleward shift (see also Fig. 12b). This picture is modified when the critical level changes occur at phase speeds where there is little wave dissipation in the mean state. For example, the U perturbation at latitude 31.5° (Fig. 13e) gives rise to a UV response that is almost exclusively present only on the equatorward flank of the imposed U perturbation. The poleward flank of the U perturbation lacks a UV response because the associated critical level changes occur at high phase speeds (18-20 m/s) where there is almost no wave activity in the mean state. This truncated UV profile causes the critical level response to project on the positive feedback (Fig. 9a). A similar situation happens for U anomalies in the tropics (Fig. 13a) except the UV response is biased to the poleward side of U and a negative feedback occurs.

To quantify the critical level effect in the simple model consider an equatorward wave activity flux, F, from distant sources. Because the sources are distant, knowing the dependence of the wave

activity flux on c alone completely characterizes the effect of the critical level. The momentum flux at a given latitude is given by the integral of F up to the critical line at the latitude:

$$\overline{u'v'} = \int_{-\infty}^{\overline{u}} F(c)dc, \tag{17}$$

where we assume cartesian geometry for simplicity. The change in $\overline{u'v'}$ per $\overline{u}$ is therefore:

$$\frac{d\overline{u'v'}}{d\overline{u}} = F(\overline{u}). \tag{18}$$

Writing F in terms of the latitude by defining $G(\phi) \equiv F(\overline{u}(\phi))$ and rearranging:

$$\Delta\overline{u'v'} = G(\phi)\Delta\overline{u}, \tag{19}$$

where Δ denotes anomalies in eddy momentum flux and U. The function G is the amount of wave dissipation as a function of ϕ in the climatology. When U is centered over the maximum in G, UV is shaped like U (see Fig. 13c). In this case, the eddies cause a pure poleward shift because EMFC is out of phase with UV. As the U shifts off the G distribution, the UV profile is truncated (Fig. 13ae), and a portion the EMFC leads to either positive or negative feedback (see the red line in Fig. 9a).

It is also noteworthy that the critical level effect (19) does not depend on the meridional shear of the mean U despite the fact that smaller shear means the critical level moves further per U. It turns out smaller shear also spans less range in c and therefore less of the wave activity flux, F, is involved. In the simple model, these effects exactly cancel (see also Lorenz (2014b)).

To understand the coupled interactions that govern the $\overline{u}$ anomalies, (19) can be combined with the vertically averaged momentum budget:

$$\frac{\partial\overline{u}}{\partial t} = -\frac{\partial\overline{u'v'}}{\partial y} = -\frac{\partial G(y)\overline{u}}{\partial y}, \tag{20}$$

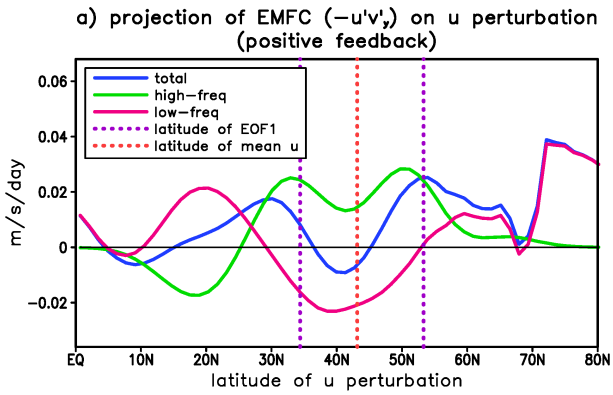

FIG. 14. a) Positive feedback of the EMFC from the perturbed simple model onto the imposed Gaussian U anomaly from all eddies (blue), the high frequency eddies (green) and the low frequency eddies (magenta). The latitudes of the time-mean jet (red) and the centers of action of EOF1 (purple) are also shown.

where we drop the Δ symbol for the anomalies. Therefore U under critical level dynamics obeys a simple conservation equation with transport velocity G (Lorenz (2014b)):

$$\frac{\partial \overline{u}}{\partial t} + \frac{\partial G(y)\overline{u}}{\partial y} = 0. \tag{21}$$

Therefore critical level dynamics is always causing the "center of mass" of the U profile to migrate towards the wave source. The continuous latitude shift of U anomalies with critical level dynamics is very distinct from the reflection case where there is a fixed point that attracts all U anomalies.

e. High versus Low Frequency Eddies

It has been known for a long time that the high frequency eddies are responsible for the positive feedback and that the low frequency eddies actually damp the dominant U EOF (Robinson (1991); Lorenz and Hartmann (2001); Lorenz and Hartmann (2003)). The simple model reproduces this behavior (Fig. 14), where we use a period of 10 days to separate the high and low frequency eddies. Here we show the projection of EMFC back on the imposed U perturbation (see section 3b) for a range of different latitudes. The high frequency eddies (green) are responsible for the positive at the centers of action of EOF1 (purple dotted lines). The low frequency eddies (magenta) have a negative feedback for the equatorward center of action. It is sometimes argued that the high frequency eddies cause the positive feedback because they are baroclinic waves and that the

latitude of baroclinic instability shifts with the jet. In this model, however, the amplitude of the baroclinic source is fixed. Instead, the high frequency feedback is due to two factors. 1) For the poleward center of action, reflection dynamics are at play. Changes in wave reflection occur at phase speeds near the peak of the reflecting level (Fig. 10). The peaks in the reflecting level for the dominant zonal wavenumbers 5-8 correspond to wave periods of 7.5, 5.3, 3.9 and 3.1 days, respectively. All these periods are in the high frequency range and so the positive feedback due changes in reflection comes from the high frequency eddies. 2) For the equatorward center of action, critical level dynamics are at play. For the critical level, the positive feedback occurs when U is at the poleward margin of G (19), otherwise UV acts to shift U poleward rather than reinforce (see section 3d). The poleward margin of G (i.e. the wave dissipation zone on the equatorward flank of the jet) is due to the high frequency eddies as well. For example, compare the phase speeds of the UV response in Fig. 13f with those in Fig. 13bd.

f. Observations

The structures of poleward, equatorward and stationary U anomalies seen in the GCM and simple model are also seen in observations. In Fig. 15, we look at poleward propagation and persistence (see section 2f) over the annual cycle in observations from the JRA-55 reanalysis (Kobayashi et al. (2015); Harada et al. (2016)) from 1958-2017. To minimize the effect of subtropical jets, U is averaged from 500-1000mb. As seen in the simple model (Fig. 8b), there are two distinct regions of poleward propagation in the Southern Hemisphere throughout the year: one in the subtropics and another centered at the latitude of the mean 500-1000mb mid-latitude jet (dotted black line, Fig. 15a). Also, like the simple model, there is equatorward propagation further poleward and therefore the latitudes on the poleward flank of the jet (60°S) attract U anomalies. In the Northern Hemisphere, the same pattern holds for January through March, however, in the other seasons there is a third region of poleward propagation even further poleward. Also, the alignment between a relative maximum in poleward propagation and the mean jet disappears in the latter half of the year. Instead, there is a broad latitude zone of poleward propagation. Looking at the persistence (Fig. 15b), the Southern Hemisphere jet in summer has zones of increased persistence on each flank of the mid-latitude just like the simple model. In Southern Hemisphere winter, however, the equatorward zone of persistence migrates further equatorward. Since this center of action is reinforced by

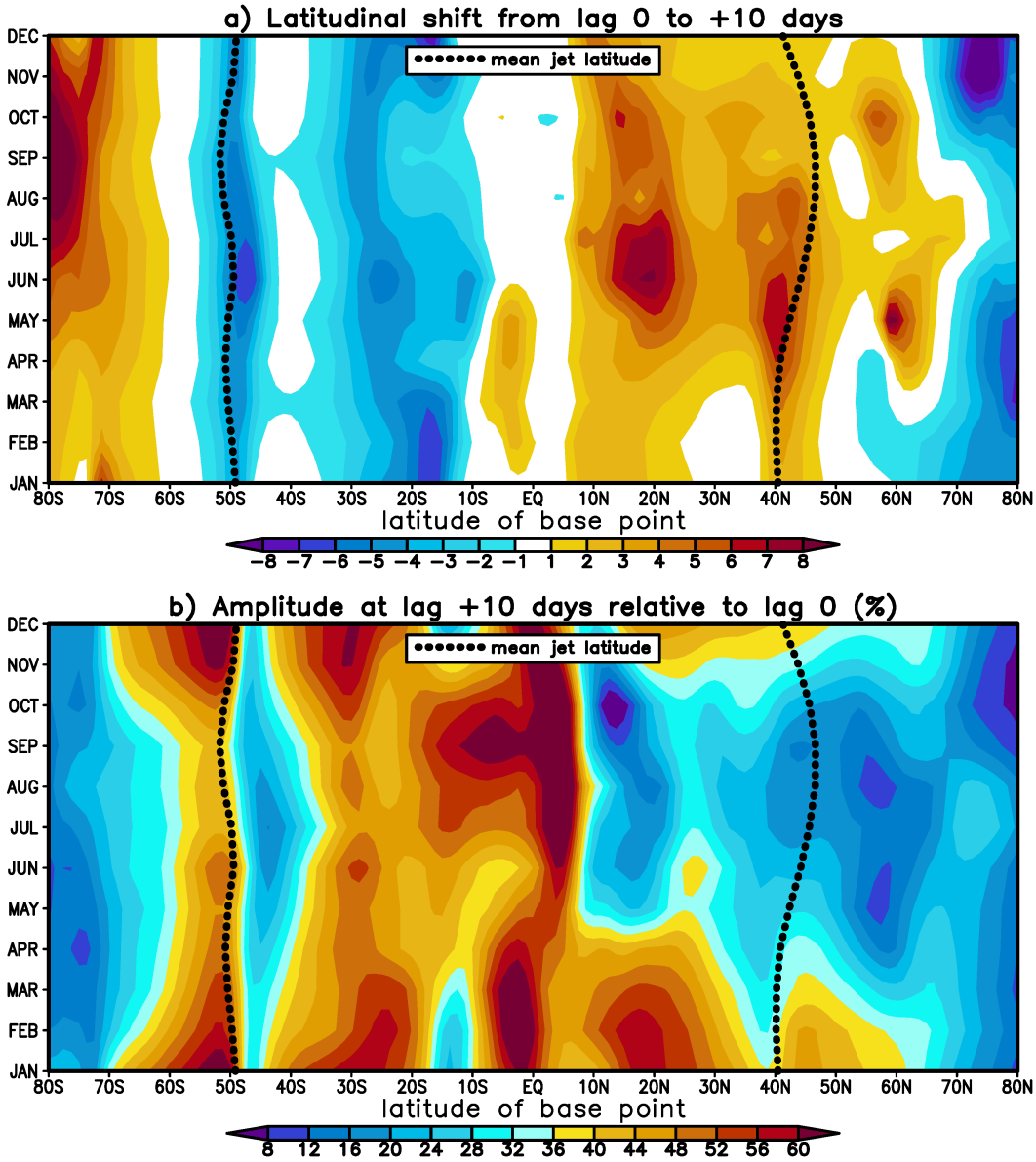

FIG. 15. a) Estimated latitude shift from lag 0 to lag 10 days in observations estimated from one-point-lagged regressions (shaded, units = latitude/(10 days) (see (10) in section 2diagnostics). The latitude shift is shown as a function of latitude of the base point (x-axis) and time of year (y-axis). Analysis for each month is performed over the 3 month time period centered on the given month. The latitude of the time-mean jet is given by the thick dotted line. b) Same as (a) but for the amplitude at lag 10 relative to lag 0 (dimensionless). This is a measure of persistence.

critical level dynamics in the simple model, the observed migration might be consistent with the equatorward shift of the critical levels due to the strong subtropical jet in Southern Hemisphere

winter. In Northern Hemisphere winter the relationship between persistence and mean jet latitude is the same as the simple model. By late summer and fall, however, the distinct zones of excess persistence become unfocused and in some cases disappear. Overall, the behavior of the Southern Hemisphere jet and the Northern Hemisphere winter jet is remarkably similar to that in the simple model.

4. Summary

We developed a simple mechanistic model of the effect of U on eddy momentum fluxes based on the linearized nondivergent barotropic vorticity equation. The background zonal-mean state only enters the dynamics via the critical level and the reflecting level. In other words, the model is a rudimentary form of ray tracing where the full Index of Refraction (IOR) is distilled to the critical level (IOR = ∞) and the reflecting level (IOR = 0). The only other input to the model is the latitude-wavenumber-phase speed spectrum of the baroclinic wave source (i.e. convergence of the vertical EP flux in the upper troposphere). The model is used to understand the mechanisms of positive eddy-zonal-flow feedbacks and the poleward and equatorward propagation of U anomalies.

The effect of the reflecting level on the eddies is dominated by changes in the peak phase speed of the reflecting level, which for spherical geometry occurs on the poleward flank of the jet. Increases in U and/or decreases in barotropic PV gradient in this region increase the peak of the reflecting level and therefore increase the range of eddy phase speeds that reflect on the poleward flank of the jet. The key to understanding the coupled eddy-zonal-flow response is to note that the shape of UV response to reflection is insensitive to the latitude of the U anomaly as long as the anomaly is at or poleward of mean jet. For positive U, this reflection induced UV profile always accelerates the winds on the poleward flank of the jet (57° latitude in our simulation) and therefore any U anomaly at or poleward of mean jet is "attracted" toward 57°. Not coincidentally, the poleward center of action of the annular mode is close to this attracting latitude. This reflection induced attractor is also the reason stronger jets shift poleward.

The momentum flux response from critical level dynamics is proportional to the product of the U anomaly and the latitudinal profile of the (critical-level induced) dissipation in the mean state. Therefore, where the latitudinal profile of dissipation is relatively constant, UV is proportional to U and therefore the EMFC is orthogonal to U. Thus critical level dynamics leads to U propagation

toward the wave source. The U anomaly propagation speed maximizes in the subtropics where wave dissipation is largest. When the U anomaly is positioned on the source side of the peak of the mean dissipation profile, the UV profile is truncated on the source side and therefore a portion of the critical level response leads to a positive feedback. Similarly, U anomalies positioned on the other side of the mean dissipation have a negative feedback. However, unlike reflecting level dynamics, critical level dynamics is always accompanied by some degree of propagation toward the wave source.

The simple model also predicts that the high-frequency eddies are responsible for the positive eddy feedback (Robinson (1991); Lorenz and Hartmann (2001)). For reflecting level dynamics, the positive feedback involves waves with phase speeds of the peak of the reflecting level. For the dominant zonal wavenumbers, these phase speeds are in the "high-frequency" range. For critical level dynamics, the positive feedback involves the waves that dissipate closest to the jet core (i.e. the source side of the mean wave dissipation: see previous paragraph). These waves also have relatively high phase speeds because they reach their critical level sooner.

In observations, the patterns of poleward/equatorward propagation and U persistence are remarkably similar to the GCM and the simple model. In future work, we will apply the simple model to a range of different mean states to understand the robust and non-robust aspects of the eddy feedback. In particular, mean states where the subtropical jet dominates over mid-latitude jet potentially have different dynamics (Barnes and Hartmann (2011)).

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Data availability statement. Please contact the author for the model code and data.

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